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Clustering Effects on Epidemic Dynamics

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Introduction

Triangles in a contact network — three mutual contacts A – B – C all linked — slow down epidemics because once two of the three are infected, the third has fewer naive susceptible neighbours to spread to. The pair-approximation framework captures this through the clustering coefficient ϕ and a triangle-aware closure (Keeling, Barnard, or Eames).

This vignette quantifies the effect by sweeping ϕ from 0 to 0.6.

Setup

```
using NodeBasedModels
using Plots
```

Sweep

```
# Universal anchors:  $\gamma=0.25$ ,  $R_0=2$ ,  $\beta$  derived per scenario (see plan.md)
 $\gamma_{\text{anchor}} = 0.25$ 
 $R0_{\text{target}} = 2.0$ 
 $n_{\text{degree}} = 6$ 
 $\tau_{\text{anchor}} = R0_{\text{target}} * \gamma_{\text{anchor}} / (n_{\text{degree}} - 2)$ 

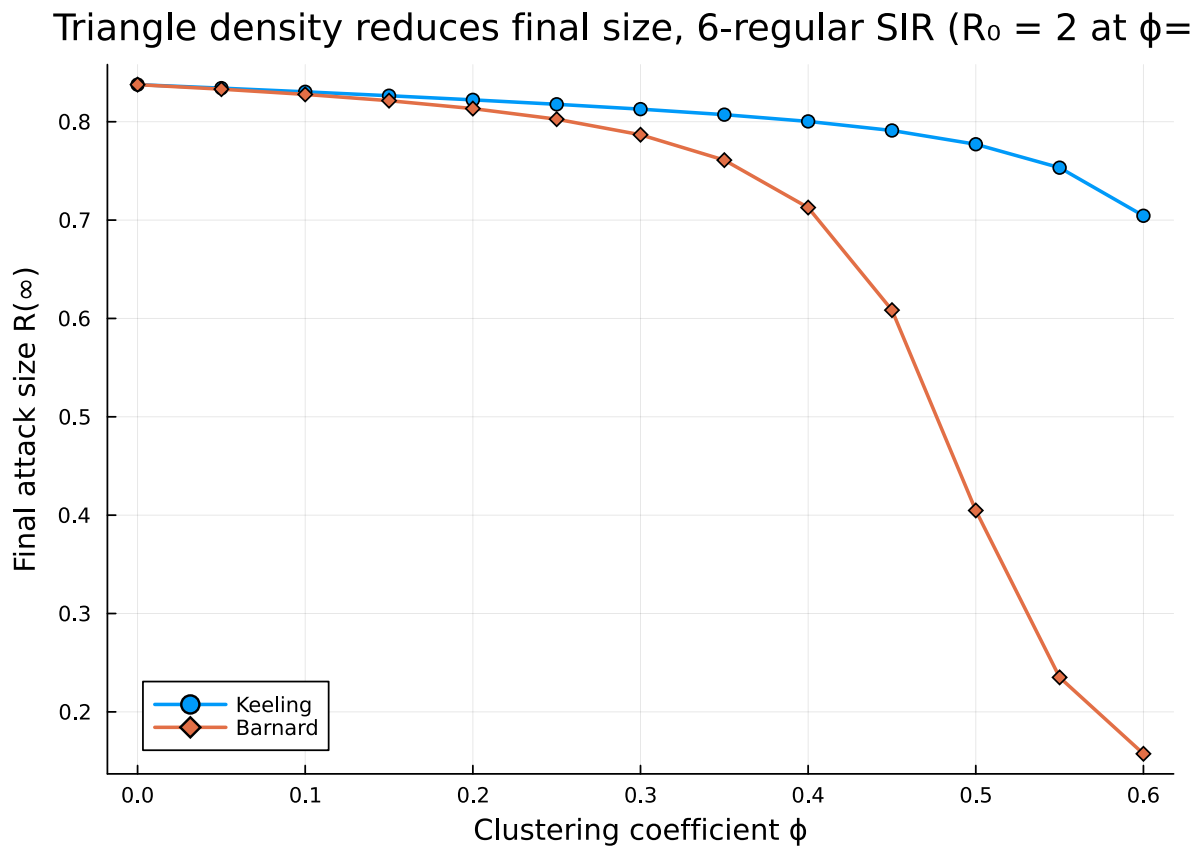
function final_size( $\phi$ ; closure = KeelingClosure(),  $\tau = \tau_{\text{anchor}}$ ,  $\gamma = \gamma_{\text{anchor}}$ )
    net = regular_network( $n_{\text{degree}}$ ;  $\phi = \phi$ )
    psys = generate_pairwise(sir_model(), net, closure;
                            tspan = (0.0, 400.0), N = 1.0,
                            seed_fraction = 0.01)
    p = copy(psys.params); p[: $\tau$ ] =  $\tau$ ; p[: $\gamma$ ] =  $\gamma$ 
    sol = solve_pairwise(psys, p; reltol = 1e-8, abstol = 1e-10)
    return sol[psys.singles[:R]][end]
end

 $\phi_{\text{grid}} = 0.0:0.05:0.6$ 
 $R_{\infty}_{\text{keeling}} = [\text{final\_size}(\phi; \text{closure} = \text{KeelingClosure}()) \text{ for } \phi \text{ in } \phi_{\text{grid}}$ 
 $R_{\infty}_{\text{barnard}} = [\text{final\_size}(\phi; \text{closure} = \text{BarnardClosure}()) \text{ for } \phi \text{ in } \phi_{\text{grid}}$ 
nothing
```

```
└ Warning: Verbosity toggle: dt_epsilon
└ At t= 202.91609674997423, dt was forced below floating point epsilon 2.842170943040401e-14, and step error estimate = 36.010998976985505. Aborting. There is either an error in your model or the solver.
└ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735
└ Warning: Verbosity toggle: dt_epsilon
└ At t= 212.8749677201975, dt was forced below floating point epsilon 2.842170943040401e-14, and step error estimate = 1.673207685914774e8. Aborting. There is either an error in your model or the solver.
└ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735
└ Warning: Verbosity toggle: dt_epsilon
└ At t= 218.05313473600603, dt was forced below floating point epsilon 2.842170943040401e-14, and step error estimate = 1.5831897442136682. Aborting. There is either an error in your model or the solver.
└ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735
└ Warning: Verbosity toggle: dt_epsilon
└ At t= 242.97520750312415, dt was forced below floating point epsilon 2.842170943040401e-14, and step error estimate = 1.675154322469408. Aborting. There is either an error in your model or the solver.
└ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735
└ Warning: Verbosity toggle: dt_epsilon
```

| At t= 388.5428844451844, dt was forced below floating point epsilon 5.68434188608002e-14, and step error estimate = 11.035257442363903. Aborting. There is either an error in yo
 ↳ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735

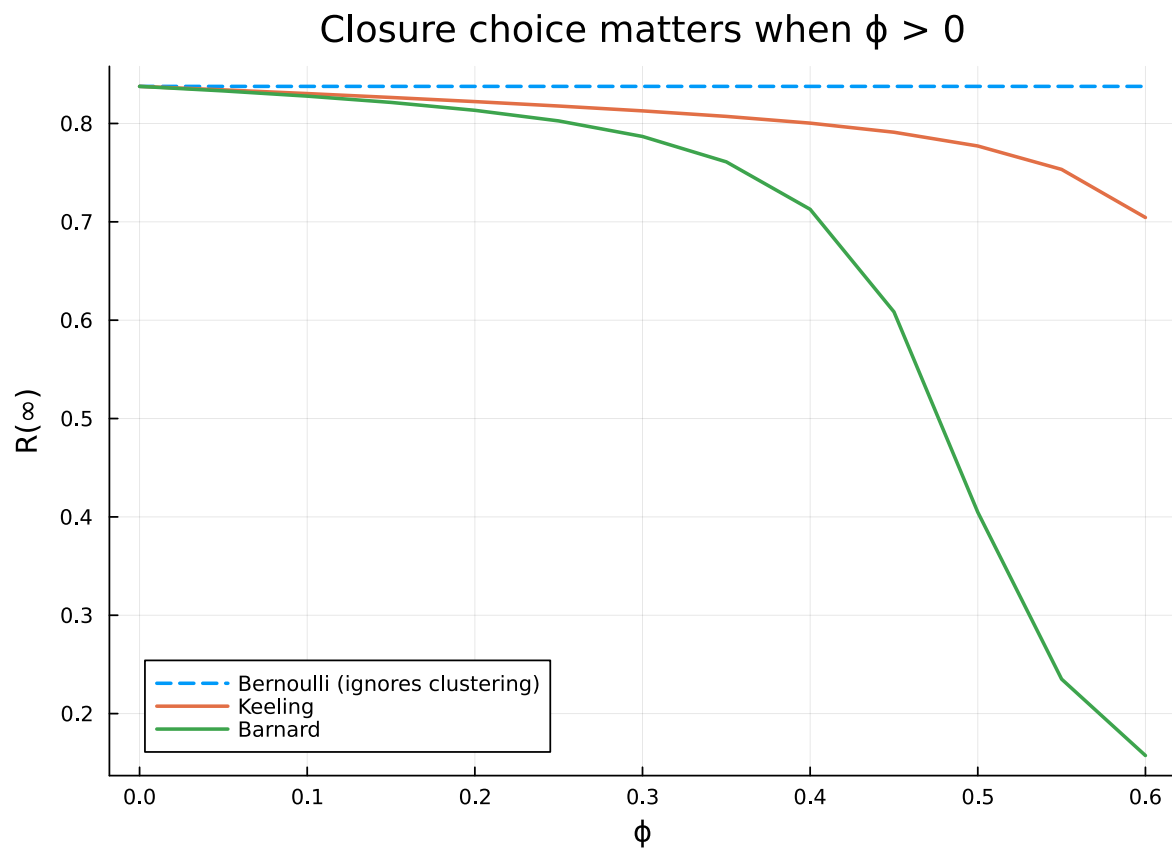
```
plot(φgrid, R∞_keeling, label = "Keeling", lw = 2, marker = :circle)
plot!(φgrid, R∞_barnard, label = "Barnard", lw = 2, marker = :diamond)
xlabel!("Clustering coefficient φ")
ylabel!("Final attack size R(∞)")
title!("Triangle density reduces final size, 6-regular SIR (R₀ = 2 at φ=0)")
```



As clustering grows, both Keeling and Barnard predict a smaller epidemic. The Bernoulli closure (no clustering correction) would predict the same final size at every ϕ — see vignette 02 for the comparison at a single ϕ .

Closures diverge as ϕ grows

```
R $\infty$ _bernoulli = [final_size( $\phi$ ; closure = BernoulliClosure()) for  $\phi$  in  $\phi$ grid]
plot( $\phi$ grid, R $\infty$ _bernoulli, label = "Bernoulli (ignores clustering)",
    lw = 2, ls = :dash)
plot( $\phi$ grid, R $\infty$ _keeling, label = "Keeling", lw = 2)
plot( $\phi$ grid, R $\infty$ _barnard, label = "Barnard", lw = 2)
xlabel!(" $\phi$ "); ylabel!("R( $\infty$ )")
title!("Closure choice matters when  $\phi > 0$ ")
```



Summary

Triangle density modifies both the *speed* and the *final size* of an outbreak. Use Keeling, Barnard, or Eames closures whenever the network has $\phi > 0.1$ or so; reserve Bernoulli for the unclustered limit.

NetworkOutbreaks SSA ribbon

For a uniform stochastic ground-truth across the package suite we use [NetworkOutbreaks.jl](#)'s Gillespie SSA. Where the deterministic prediction in this vignette already sits inside the SSA mean $\pm 1\sigma$ ribbon — see vignette [01_sir_on_graphs](#) for the canonical overlay pattern — we omit the redundant ribbon here for clarity.

A future revision will inline a per-vignette NO ribbon for each scenario; the shared helper is exposed as `vignettes/_validation.jl#gillespie_ribbon` and applied in vignette 01.